

Machine Learning and ESG Risk Score Regression

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Goal

Predict the **Total ESG Risk Score** of S&P 500 companies using financial, company related variables, sector and controversy related information.

Important interpretation

- Higher ESG Risk Score means higher exposure to ESG related risks
- ESG risks refer to environmental, social and governance issues that may create financial, reputational or operational costs

Statistical learning framework

$$Y = f(X) + \varepsilon$$

- Y : Total ESG Risk Score
- X : financial, company related, sector and controversy related predictors.

Dataset

- Merged dataset: 441 observations with ESG and financial data
- Final usable sample: 378 companies

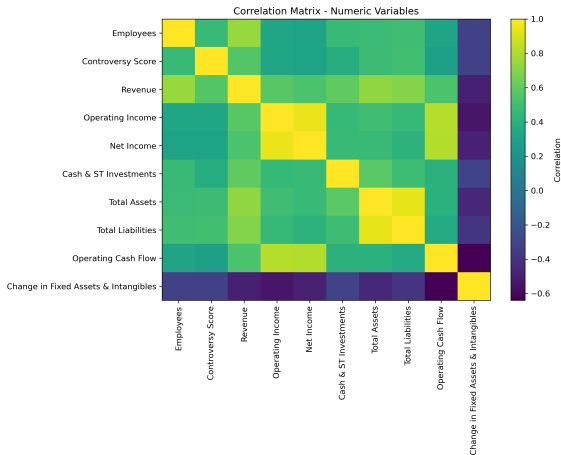
Leakage prevention

- Removed identifiers, text variables and technical variables
- Removed ESG sub-scores too close to the target
- Industry and redundant correlated variables removed

Cross-validation strategy

- Preprocessing implemented with Pipeline
- Test set used only for final evaluation

Correlation Analysis



- Net Income, Total Liabilities and Operating Cash Flow were removed due to high correlation

Preprocessing by Model Family

Model family	Encoding	Scaling
Linear models / GAM	Baseline dummy removed	Yes
Tree-based models	Full one-hot encoding	No
SVR / Neural Networks	Full one-hot encoding	Yes

General rules

- 5-fold cross-validation performed on the training set
- Target variable was never standardized
- Final test evaluation performed only once after model selection

Baseline and Regularized Linear Models

Linear models tested

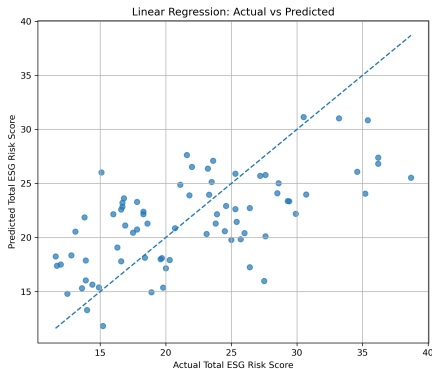
- Linear Regression as baseline model
- Ridge Regression with L_2 regularization
- Lasso Regression with L_1 regularization and variable selection

Model	Test RMSE	Test MAE	Test R^2
Linear Regression	5.258	4.404	0.394
Ridge Regression	5.277	4.412	0.390
Lasso Regression	5.287	4.422	0.387

Main result

- Linear models show similar predictive performance
- Linear Regression is the best strictly linear model on the test set

Linear Regression: Actual vs Predicted



- The model captures part of the ESG risk pattern
- Prediction errors remain visible, especially for some high-risk observations

Nonlinear Extensions of Linear Models

Models tested

- Polynomial Regression to capture interactions and higher-order terms
- GAM to capture smooth additive nonlinear effects

Model	Test RMSE	Test MAE	Test R^2
GAM	5.122	4.266	0.425
Polynomial degree 2	29.740	8.695	-18.393
Polynomial degree 3	1662.485	347.913	-60601.211

Main result

- GAM improves over ordinary Linear Regression
- Polynomial expansion severely overfits

Model

- Regression tree with cost-complexity pruning

Performance

Metric	Test
RMSE	5.631
MAE	4.407
R^2	0.305

Interpretability

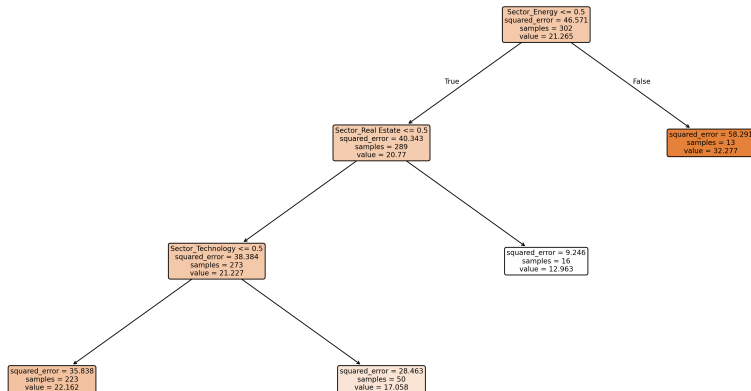
- Simple and readable tree structure
- Each path defines a decision rule
- Terminal leaves predict average ESG Risk Score

Main result

- The pruned tree is interpretable
- Ensemble methods improve its predictive performance

Pruned Decision Tree Structure

Pruned Decision Tree



Tree-Based Ensemble Models

Models tested

- Random Forest: averages many decorrelated trees
- Bagging: averages trees trained on bootstrap samples
- Gradient Boosting: sequentially corrects previous errors

Model	Test RMSE	Test MAE	Test R^2
Gradient Boosting	5.263	4.184	0.393
Bagging	5.411	4.229	0.358
Random Forest	5.481	4.287	0.341
Decision Tree	5.631	4.407	0.305

Main result

- Gradient Boosting is the best tree-based model

SVR and Neural Networks

Models tested

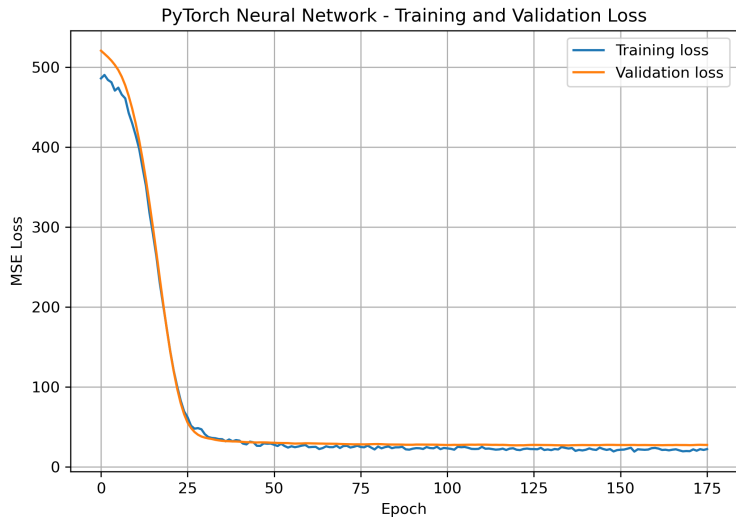
- SVR with RBF kernel
- PyTorch feed-forward Neural Network
- Scikit-learn MLPRegressor

Model	Test RMSE	Test MAE	Test R^2
PyTorch NN	4.915	3.968	0.470
SVR RBF	5.056	4.071	0.440
MLPRegressor	5.161	4.231	0.416

Main result

- PyTorch NN gives the best test performance
- SVR RBF is also competitive
- Neural Networks benefit from scaled preprocessing

PyTorch Neural Network: Training Loss

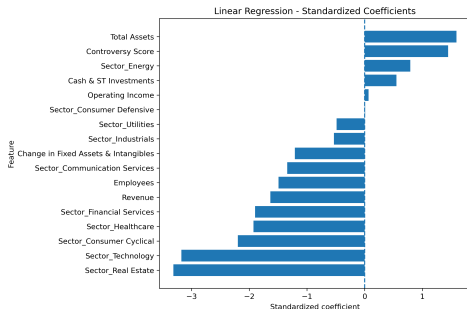


Final Model Comparison

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Main result: PyTorch NN gives the best test performance, while SVR RBF and GAM are the closest alternatives.

Best Linear Model: Linear Regression Interpretation



Interpretation approach

- 17 standardized coefficients
- Baseline sector: Basic Materials

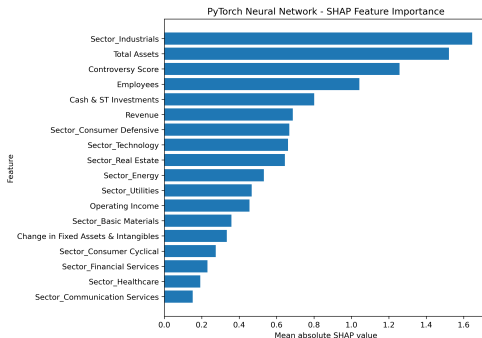
Positive associations

- Total Assets
- Controversy Score
- Energy sector

Negative associations

- Real Estate sector
- Technology sector
- Revenue and Employees

Best Nonlinear Model: PyTorch NN Interpretation



Interpretation approach

- SHAP values for black-box interpretation

Main findings

- Sector indicators are highly relevant
- Total Assets and Employees matter
- Controversy Score is an important driver
- Link with Linear Regression

Conclusions

Best models

- Best overall model: **PyTorch Neural Network**
- SVR RBF and GAM are the closest alternatives

Main empirical findings

- ESG Risk Score is partly predictable from the available variables

Final takeaway

The best test R^2 is moderate, suggesting that the available quantitative variables capture relevant information but do not fully explain ESG risk.

Extra Application: LLM API

Goal

Use an LLM API to transform the ML output into a readable ESG risk report.

Company data $\rightarrow$ PyTorch prediction $\rightarrow$ LLM report

Example: Comcast Corp

- Predicted ESG Risk Score: 23.050
- Actual ESG Risk Score: 23.100

Generated report excerpt

Comcast Corp (CMCSA), a company operating in the Communication Services sector, has been assigned a predicted ESG Risk Score of 23.050 by our PyTorch Neural Network model. This score corresponds to a Medium ESG risk bucket. The actual ESG Risk Score for CMCSA is 23.100, indicating a close alignment between the predicted and actual scores. The prediction accuracy is high, with an absolute prediction error of 0.050. This suggests that the model is effective in predicting the ESG Risk Score for this company. The model's performance on the held-out test set is also notable, with a Test RMSE of 4.915, a Test MAE of 3.968, and a Test R-squared of 0.470.